# CMS-0042-NC Full Comment Outline – Blockchain-Enabled Health Ecosystem (Goliath Engineering Technology)

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## I. Executive Summary

CMS has taken important steps toward modernizing healthcare interoperability and empowering patient access through initiatives such as Blue Button 2.0, TEFCA, and the Interoperability and Patient Access Final Rule.
However, **without a secure, patient-centered, blockchain-enabled infrastructure**, many of the core challenges — fragmented data, inefficient processes, fraud, and compliance risk — will persist.

**Goliath Engineering Technology** offers a **blockchain-based health information framework** that directly addresses these gaps and accelerates CMS's health technology vision:

- Real-time, **patient-controlled** health data access
- Seamless, **secure interoperability** across systems
- Trusted, **decentralized digital identity management**
- Immutable **audit trails** ensuring compliance
- **Fraud, waste, and abuse prevention** at the point of transaction
- Frictionless enablement of **value-based care (VBC)** workflows

Our solution is ready for rapid deployment and scalable across the CMS beneficiary population.

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## II. Core Problems Blockchain Directly Solves (Tie to CMS RFI Questions)

| CMS RFI Area | Blockchain Solution (Goliath Engineering) |
|---|---|
| **PC-2, PC-8, PC-9:** Patient data fragmentation. | Goliath's blockchain EHR wallet aggregates all patient-authorized data into one secure, portable record. |
| **PC-14, PR-9, PR-10, PA-3:** Identity and credentialing inconsistency. | Decentralized self-sovereign identity (SSI) allows a patient to use one credential across providers and systems. |

| CMS RFI Area | Blockchain Solution (Goliath Engineering) |
|---|---|
| **PR-5, PR-7, TD-4, TD-5:** Inconsistent interoperability across APIs, HIEs, TEFCA. | Goliath's blockchain trust layer standardizes and secures data exchange across systems. |
| **PC-13, PR-12, TD-18:** Information blocking issues. | Smart contracts automate permissioned data sharing and provide immutable audit logs. |
| **VB-1, VB-3, VB-4:** Value-based care (VBC) data gaps. | Blockchain enables trusted, real-time outcomes reporting and secure care coordination. |
| **TD-1, TD-2, TD-7:** Innovation bottlenecks. | Goliath's blockchain framework provides developers with secure, patient-permissioned access to CMS and clinical datasets. |

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# III. Proposed Pilot Model: Goliath Blockchain Health Wallet Initiative

CMS can immediately initiate a **blockchain health wallet pilot** via Goliath Engineering Technology by:

- **Integrating Blue Button 2.0 APIs** into a blockchain EHR wallet
- **Enabling patients** to control and share Medicare claims, clinical data, and appointment navigation records
- **Deploying decentralized digital identity** solutions compliant with NIST 800-63-3 and compatible with Login.gov frameworks
- **Implementing smart contract-based consent** for dynamic, real-time data sharing permission management
- **Testing bulk FHIR data flows** under real-world care coordination scenarios

**Expected Outcomes:**

- Full patient control and access across disparate providers
- Seamless data sharing without repeated authentication
- Real-time compliance with HIPAA, TEFCA, and CMS interoperability rules
- Enhanced provider and payer collaboration under value-based models

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# IV. Additional Strategic Benefits

- **Security & Privacy**: Blockchain-native encryption and access logs greatly exceed traditional HIPAA-compliance thresholds.
- **Data Liquidity**: Instant, standardized interoperability across CMS payers, providers, developers, and apps.

- **Cost Reduction**: Eliminates redundant integration costs and dramatically reduces portal maintenance overhead.
- **Developer Enablement**: Secure permissioned data access accelerates innovation without compromising patient control.

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# V. Fraud, Waste, and Abuse (FWA) Prevention and Estimated Savings

Goliath Engineering Technology's blockchain framework addresses healthcare's FWA crisis at the root cause.

### Capabilities:

- **Immutable Ledgering**: Every claim, service authorization, and consent action is permanently and transparently recorded.
- **Smart Contract Validation**: Eligibility, authorization, and service verification occur *before* billing submission.
- **Decentralized Identity Management**: Stops ghost billing, duplicate claims, and unauthorized services by tying claims to verified identities.

### Estimated Financial Impact:

- **Current Medicare improper payment estimate**: ~$30 billion/year
- **Projected savings**: Even a **10% reduction** via blockchain validation yields **$3 billion annual savings**.
- **At full system scale (Medicare + Medicaid)**: Projected annual systemic savings of **$10–$15 billion**.

Blockchain prevents fraud at the **point of transaction**, rather than relying on costly and delayed "pay and chase" recoveries.

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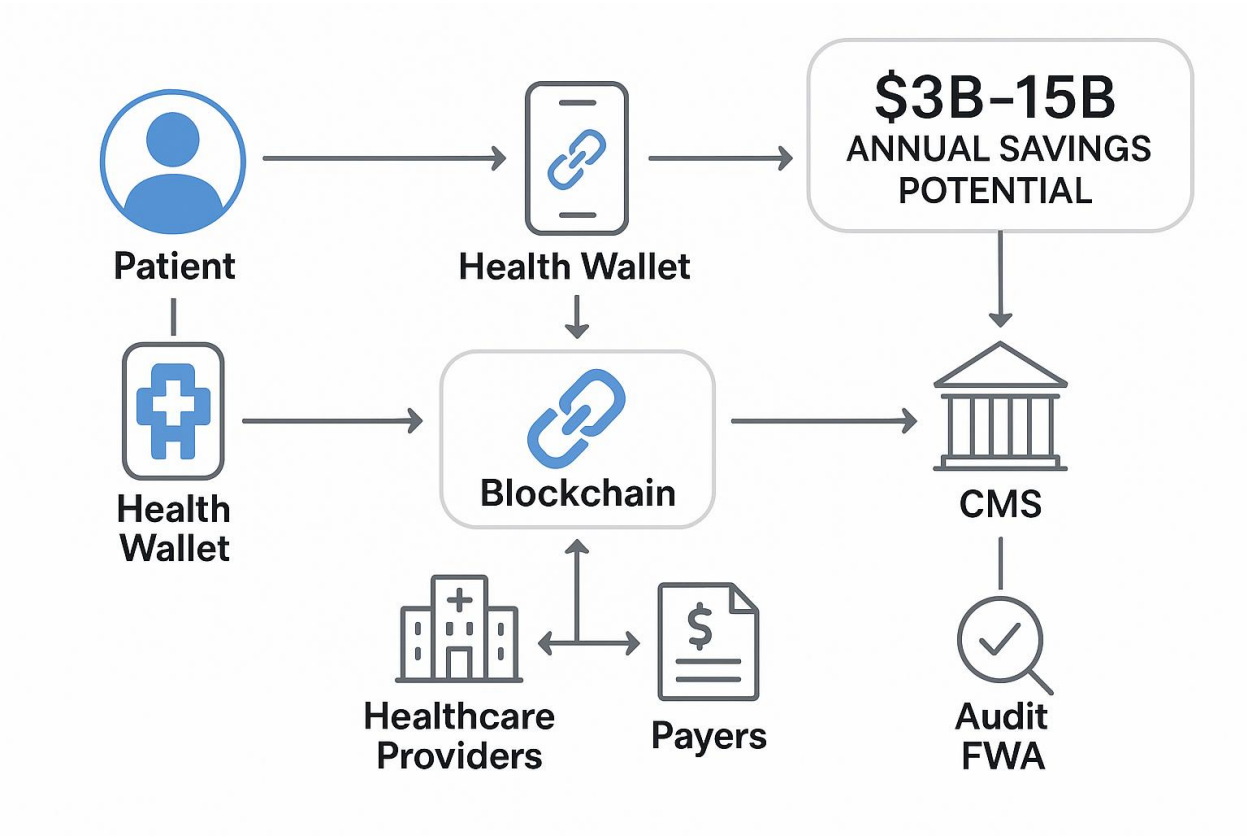
# VI. CMS-Specific Requests for Action

We respectfully recommend CMS:

1. **Launch a Goliath blockchain health wallet pilot** integrated with Blue Button 2.0 APIs.
2. **Implement decentralized identity credentialing** to unify patient access across providers.
3. **Adopt blockchain-based consent management models** within TEFCA and FHIR frameworks.

4. **Encourage pilots targeting fraud, waste, and abuse prevention** through blockchain-based transaction validation.
5. **Incentivize interoperability and real-time auditability** for advancing value-based care models and health outcomes transparency.




**Goliath Engineering Technology's Blockchain Health Ecosystem**

This diagram illustrates a **patient-centered, blockchain-enabled health data framework** aligned with CMS's interoperability and fraud reduction objectives.

- **Patient-Controlled Health Wallets** aggregate and manage Medicare claims, clinical records, and care navigation information.
- **Blockchain Infrastructure** ensures real-time, standards-based, permissioned interoperability across healthcare providers, payers, and CMS systems.
- **Immutable Ledgering and Smart Contract Validation** prevent fraud, waste, and abuse at the point of service, significantly reducing improper payments.
- **Projected Savings**: Conservative modeling indicates **$3–$15 billion annually** in systemic savings by reducing improper billing and enhancing data trustworthiness.
- **Alignment with CMS Goals**: Fully supports interoperability mandates (FHIR, TEFCA, Blue Button 2.0) and value-based care (VBC) initiatives.

This integrated, scalable solution delivers **greater patient empowerment, operational efficiency, financial stewardship, and compliance assurance** — immediately actionable through a CMS-sponsored pilot.